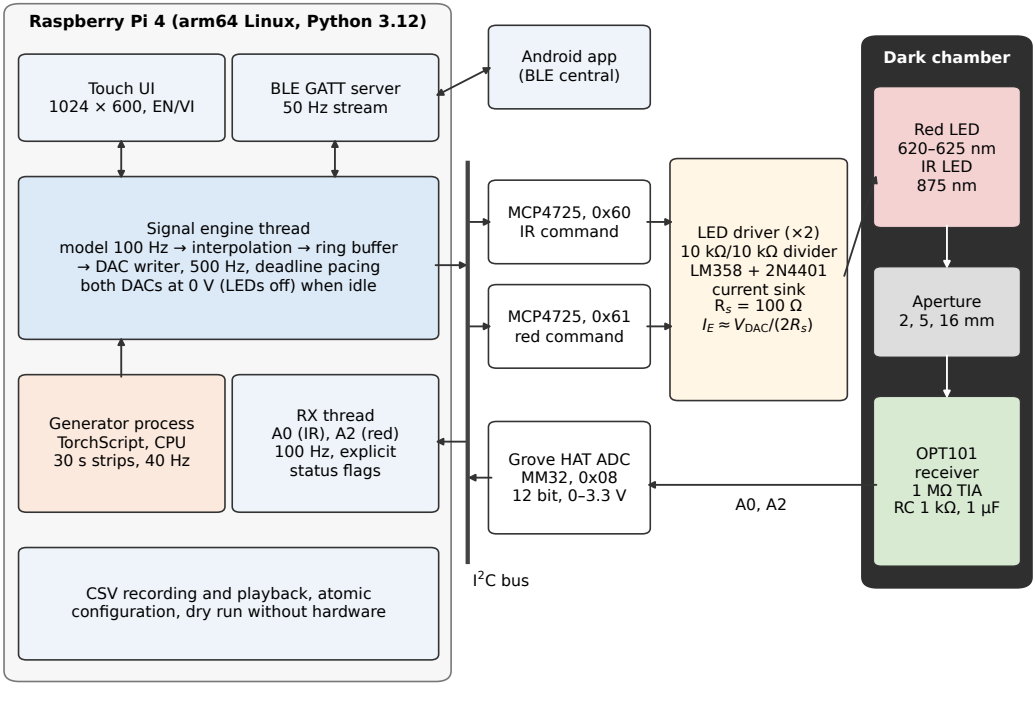
Raspberry Pi 4 (arm64 Linux, Python 3.12)
Touch UI
1024 × 600, EN/VI
BLE GATT server
50 Hz stream
Android app
(BLE central)
Signal engine thread
model 100 Hz → interpolation → ring buffer
→ DAC writer, 500 Hz, deadline pacing
both DACs at 0 V (LEDs off) when idle
Generator process
TorchScript, CPU
30 s strips, 40 Hz
RX thread
A0 (IR), A2 (red)
100 Hz, explicit
status flags
CSV recording and playback, atomic
configuration, dry run without hardware
MCP4725, 0x60
IR command
MCP4725, 0x61
red command
LED driver (×2)
10 kΩ/10 kΩ divider
LM358 + 2N4401
current sink
$R_s$ = 100 Ω
$I_E \approx V_{DAC}/(2R_s)$
Grove HAT ADC
MM32, 0x08
12 bit, 0–3.3 V
$I^2C$ bus
Dark chamber
Red LED
620–625 nm
IR LED
875 nm
Aperture
2, 5, 16 mm
OPT101
receiver
1 MΩ TIA
RC 1 kΩ, 1 µF
A0, A2